Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech
Code: 119505

Subject: Hydrology and Water Resources Engineering

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) The hydrologic cycle is driven primarily by
- (i) Earth rotation
- (ii) Wind energy only
- (iii) Solar energy
- (iv) Groundwater flow

(b) Which rain gauge records rainfall continuously?
- (i) Symon's gauge
- (ii) Tipping bucket rain gauge
- (iii) Non-recording gauge
- (iv) Storage gauge

(c) The maximum possible precipitation for a given duration over a basin is called
- (i) Average rainfall
- (ii) Probable Maximum Precipitation
- (iii) Effective rainfall
- (iv) Design storm

(d) Isohyetal method is used for estimating
- (i) Infiltration rate
- (ii) Mean precipitation over an area
- (iii) Runoff coefficient
- (iv) Evaporation losses

(e) Which method is commonly used to estimate direct runoff from rainfall?
- (i) Thiessen polygon method
- (ii) Muskingum method
- (iii) SCS-CN method
- (iv) Blaney-Criddle method

(f) Transmissivity of an aquifer is defined as
- (i) Hydraulic conductivity per unit width
- (ii) Specific yield of aquifer
- (iii) Product of hydraulic conductivity and aquifer thickness
- (iv) Rate of infiltration

(g) Which aquifer is bounded above and below by impermeable layers?
- (i) Unconfined aquifer
- (ii) Perched aquifer
- (iii) Confined aquifer
- (iv) Semi-confined aquifer

(h) The elementary profile of a gravity dam is assumed to be
- (i) Circular
- (ii) Triangular
- (iii) Rectangular
- (iv) Trapezoidal

(i) Reservoir sedimentation primarily reduces
- (i) Dam height
- (ii) Storage capacity
- (iii) Spillway capacity
- (iv) Seepage losses

(j) Water logging mainly results due to
- (i) Excessive drainage
- (ii) Low evaporation
- (iii) High groundwater table
- (iv) Canal lining

Q.2 (a) How is mean precipitation over a catchment determined? Explain Thiessen polygon and Isohyetal methods. **[7]**

(b) What is Probable Maximum Precipitation? How are intensity-duration-frequency relationships developed? **[7]**

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Q.3 (a) Explain the process of evaporation and discuss methods used for estimation of evaporation losses. [7]

(b) What is infiltration capacity? Explain Horton's infiltration equation and infiltration indices. [7]

Q.4 (a) What is a unit hydrograph? Explain its assumptions and applications. [7]

(b) A watershed of area 60 km^2 receives rainfall of 120 mm. The curve number is 75. Compute runoff depth and runoff volume. [7]

Q.5 (a) Define Duty and Delta. Also, the duty of water for a crop is 1600 hectares/cumec and base period is 120 days. Compute delta. [7]

(b) Compare surface, sprinkler, and drip irrigation methods with their advantages and limitations. [7]

Q.6 (a) Explain forces acting on a gravity dam and discuss causes of failure of gravity dams. [7]

(b) What factors govern the selection of suitable dam site and economic height of a dam? [7]

Q.7 (a) What are canal losses? Discuss methods used for prevention of seepage losses in canals. [7]

(b) A reservoir initially had storage capacity 600 Mm^3. After 20 years, sediment deposition reduced capacity to 510 Mm^3. Determine: (i) Average annual sedimentation rate (ii) Percentage loss in storage capacity [7]

Q.8 (a) Explain the design considerations for an earthen (embankment) dam and the methods to control seepage through the body of the dam. [7]

(b) What is water logging? Explain its causes, effects, and remedial measures. [7]

Q.9 Write short note on ***any two*** of the following: [7 x 2 = 14]

(a) Canal lining – Types and advantages

(b) Kennedy's Theory

(c) Buttress Dam & Gravity Dam

(d) Crop Water Requirement

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